

Check Point Software Technologies Ltd **CHKP** ★★★★★ 30 Jul 2026 21:36, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
128.23 USD 30 Jul 2026	201.00 USD 29 Oct 2025 01:51, UTC	0.64	12.80 USD Bil 31 Jul 2026	 Narrow	 Mid Blend	Medium	Standard

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The primary analyst covering this company does not own its stock.

Reporting Currency: USD | Trading Currency: USD
Currency amounts expressed with "\$" are in U.S. dollars (USD) unless otherwise denoted.

Check Point Earnings: Next-Quarter Guidance Disappoints Again; Shares Undervalued

Analyst Note Rob Hales, CFA, Senior Equity Analyst, 31 Jul 2026

Check Point's second-quarter numbers were in line with FactSet consensus; however, third-quarter guidance was below. Shares fell 8% on July 30.

Why it matters: Product revenue declined 14%, and a similar decline is expected in the third quarter. While subscription revenue accelerated modestly to 11.6%, a slight deceleration is expected next quarter. This means a record fourth quarter will be needed to meet guidance.

► Growth in forward-looking measures remained weak. Calculated billings, current calculated billings, and remaining performance obligation were 0%, 2%, and 7%, respectively.

► On the bright side, emerging technologies (Secure Access Service Edge, email security, external risk management) are growing very fast with over 40% annual recurring revenue growth in the quarter. However, this remains a small part of the business.

The bottom line: We're maintaining our \$201 fair value estimate for narrow-moat Check Point, assuming product revenue headwinds will be contained to 2026. Shares look undervalued.

► Check Point is transforming its go-to-market function this year. This quarter, it announced plans to hire hundreds of additional salespeople. With tangible proof that changes are working still several quarters away, the market is likely bearish on long-term growth.

Coming up: 2026 guidance was maintained for revenue of \$2.77 billion-\$2.85 billion and Non-GAAP EPS of

Financial Summary and Key Statistics

	Actual		Forecast	
	2024	2025	2026	2027
Revenue (USD Mil)	2,565	2,725	2,818	3,023
Revenue Growth %	6.2	6.3	3.4	7.3
Operating Income (Mil)	876	831	902	907
Operating Margin %	34.2	30.5	32.0	30.0
Adjusted EBITDA (Mil)	972	934	1,000	1,005
Adjusted EBITDA Margin %	37.9	34.3	35.5	33.2
Earnings Per Share (Diluted) (USD)	7.46	9.61	8.24	9.15
Adjusted Earnings Per Share (Diluted) (USD)	9.16	11.89	10.11	11.30
Adjusted EPS Growth %	8.8	29.7	-15.0	11.8
Price/Earnings	20.4	15.6	12.7	11.3
Price/Book	7.6	7.1	5.7	6.7
EV/EBITDA	19.3	19.4	12.8	12.7
Free Cash Flow Yield %	—	—	—	—

Source: Morningstar Valuation Model. Data as of 30 Jul 2026.

Accounts Payable = Accounts Payable + Deferred Income

Check Point Software Technologies LtdCHKP★★★★★

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128.23 USD	201.00 USD	0.64	12.80 USD Bil	 Narrow	 Mid Blend	Medium	Standard
30 Jul 2026	29 Oct 2025 01:51, UTC		31 Jul 2026				

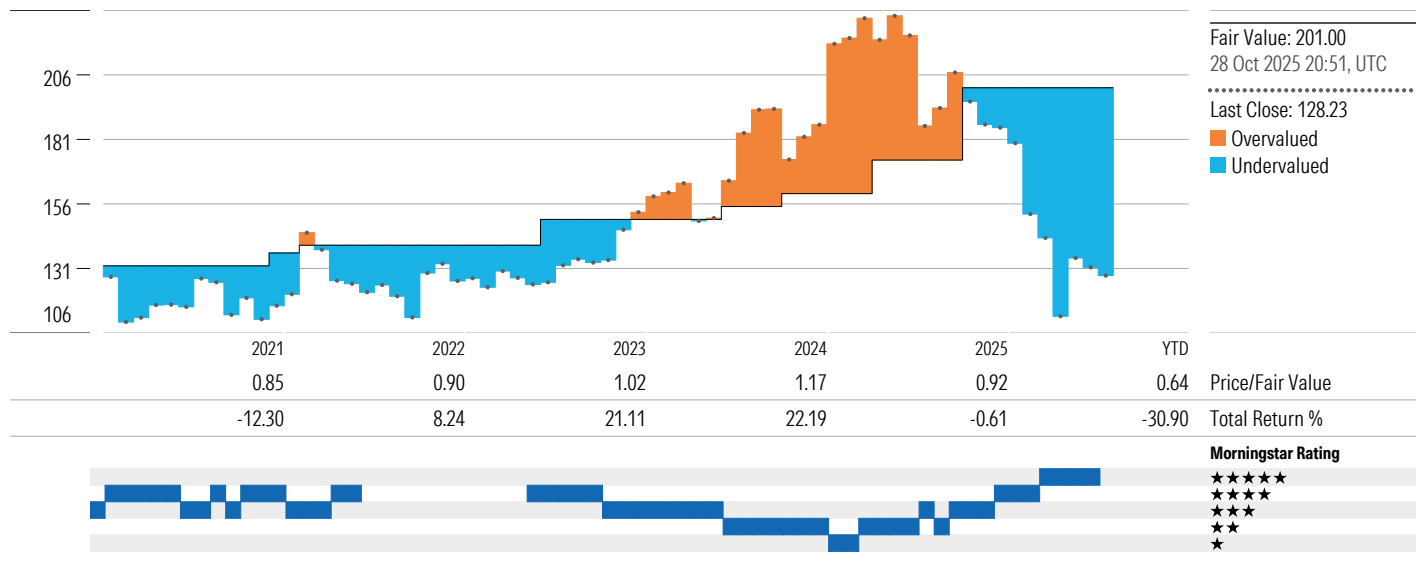
Sector	Industry	
 Technology	Software - Infrastructure	\$10.05-\$10.85.
		► Third quarter guidance is for revenue of \$655 million-\$685 million, subscription revenue of \$332 million-\$343 million, and non-GAAP EPS of \$2.43-\$2.53. 

Business Description

Check Point Software Technologies is a pure-play cybersecurity vendor. The company offers solutions for network, endpoint, cloud, and mobile security in addition to security management. Check Point, a software specialist, sells to enterprises, businesses, and consumers. Around 50% of revenue is generated in Europe, the Middle East, and Africa, 40% from the Americas, and 10% from the Asia-Pacific region. The firm, based in Tel Aviv, Israel, was founded in 1993 and has about 5,000 employees.

Check Point Software Technologies Ltd CHKP ★★★★★ 30 Jul 2026 21:36, UTC

Price vs. Fair Value



Competitors

	Check Point Software Technologies Ltd CHKP	Cisco Systems Inc CSCO	Palo Alto Networks Inc PANW	Fortinet Inc FTNT
	Fair Value 201.00 Uncertainty: Medium Last Close 128.23	Last Close 113.56 Fair Value 90.00 Uncertainty: Medium	Last Close 325.68 Fair Value 285.00 Uncertainty: Very High	Last Close 154.25 Fair Value 143.00 Uncertainty: Very High
Economic Moat	Narrow	Wide	Wide	Wide
Currency	USD	USD	USD	USD
Fair Value	201.00 30 Jul 2026 16:36, UTC	90.00 30 Jul 2026 16:34, UTC	285.00 30 Jul 2026 16:46, UTC	143.00 30 Jul 2026 16:41, UTC
1-Star Price	271.35	121.50	498.75	250.25
5-Star Price	140.70	63.00	142.50	71.50
Assessment	Undervalued 30 Jul 2026	Overvalued 30 Jul 2026	Fairly Valued 30 Jul 2026	Fairly Valued 30 Jul 2026
Morningstar Rating	★★★★★ 30 Jul 2026 05:00, UTC	★★ 30 Jul 2026 05:00, UTC	★★★ 30 Jul 2026 05:00, UTC	★★★ 30 Jul 2026 05:00, UTC
Analyst	Rob Hales, Senior Equity Analyst	William Kerwin, Senior Equity Analyst	Malik Ahmed Khan, Senior Equity Analyst	Malik Ahmed Khan, Senior Equity Analyst
Capital Allocation	Standard	Exemplary	Exemplary	Standard
Price/Fair Value	0.64	1.26	1.14	1.08
Price/Sales	4.96	7.45	22.61	15.32
Price/Book	4.87	9.16	9.57	72.86
Price/Earnings	13.12	37.85	283.20	54.51
Dividend Yield	0.00%	1.46%	0.00%	0.00%
Market Cap	13.09 Bil	447.59 Bil	265.43 Bil	113.01 Bil
52-Week Range	112.23—210.66	65.75—130.37	139.57—368.80	70.12—170.35
Investment Style	Mid Blend	Large Blend	Large Growth	Large Growth

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Morningstar Valuation Model Summary

Financials as of 30 Jul 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (USD Mil)	2,415	2,565	2,725	2,818	3,023	3,284	3,561	3,834
Operating Income (USD Mil)	899	876	831	902	907	1,014	1,128	1,244
EBITDA (USD Mil)	957	972	932	999	1,003	1,111	1,226	1,343
Adjusted EBITDA (USD Mil)	957	972	934	1,000	1,005	1,113	1,228	1,344
Net Income (USD Mil)	840	846	1,056	863	867	958	1,055	1,154
Adjusted Net Income (USD Mil)	997	1,039	1,306	1,059	1,070	1,170	1,275	1,381
Free Cash Flow To The Firm (USD Mil)	1,335	1,604	1,829	1,150	666	878	1,022	1,053
Weighted Average Diluted Shares Outstanding (Mil)	118	113	110	105	95	85	77	69
Earnings Per Share (Diluted) (USD)	7.10	7.46	9.61	8.24	9.15	11.21	13.72	16.73
Adjusted Earnings Per Share (Diluted) (USD)	8.43	9.16	11.89	10.11	11.30	13.68	16.58	20.02
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 30 Jul 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	34.0	37.2	34.2	30.5	32.0	30.0	30.9	31.7	32.5	31.4
EBITDA Margin %	—	39.6	37.9	34.2	35.5	33.2	33.9	34.4	35.0	—
Adjusted EBITDA Margin %	—	39.6	37.9	34.3	35.5	33.2	33.9	34.5	35.1	34.4
Net Margin %	35.5	34.8	33.0	38.8	30.6	28.7	29.2	29.6	30.1	29.6
Adjusted Net Margin %	43.3	41.3	40.5	47.9	37.6	35.4	35.6	35.8	36.0	36.1
Free Cash Flow To The Firm Margin %	61.6	55.3	62.6	67.1	40.8	22.0	26.7	28.7	27.5	29.2

Growth & Ratios as of 30 Jul 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	5.4	3.6	6.2	6.3	3.4	7.3	8.6	8.5	7.6	7.1
Operating Income Growth %	-2.1	1.7	-2.6	-5.2	8.6	0.5	11.8	11.3	10.3	8.4
EBITDA Growth %	0.1	2.9	1.6	-4.1	7.1	0.4	10.8	10.4	9.5	7.6
Adjusted EBITDA Growth %	0.1	2.9	1.6	-3.9	7.1	0.4	10.8	10.4	9.5	7.6
Earnings Per Share Growth %	15.1	12.6	5.0	28.8	-14.2	11.1	22.5	22.4	21.9	11.7
Adjusted Earnings Per Share Growth %	15.1	13.9	8.8	29.7	-15.0	11.8	21.1	21.2	20.8	11.7

Valuation as of 30 Jul 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earnings	18.1	20.4	15.6	12.7	11.3	9.4	7.7	6.4
Price/Sales	7.4	7.9	7.2	4.6	4.3	4.0	3.7	3.4
Price/Book	6.4	7.6	7.1	5.7	6.7	8.0	9.6	11.3
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	17.2	19.3	19.4	12.8	12.7	11.5	10.4	9.5
EV/EBIT	18.3	21.4	21.8	14.2	14.1	12.6	11.3	10.3
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 30 Jul 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	14.8	14.7	13.5	11.3	12.4	14.6	16.8	19.0
ROE %	29.8	30.3	36.6	36.8	47.9	69.9	102.9	147.9
ROIC %	21.6	19.3	17.0	16.2	17.0	17.2	17.7	17.9

Accounts Payable = Accounts Payable + Deferred Income

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Financial Leverage (Reporting Currency)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	0.0	0.0	9.0	8.5	8.6	8.9	9.1	9.4
Assets/Equity	2.0	2.1	2.7	3.2	3.8	4.8	6.1	7.8
Net Debt/EBITDA	-1.6	-1.4	-1.1	-0.9	-0.2	0.2	0.4	0.6
Total Debt/EBITDA	0.0	0.0	2.1	2.0	2.0	1.8	1.6	1.5
EBITDA/ Net Interest Expense	-12.5	-10.1	-8.2	-8.8	-8.8	-9.8	-10.8	-11.8

Forecast Revisions as of 31 Jul 2026

Prior data as of 1 May 2026	2026		2027		2028	
	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	201.00	199.03	—	—	—	—
Revenue (USD Mil)	2,818	2,859	3,023	3,067	3,284	3,329
Operating Income (USD Mil)	902	964	907	925	1,014	1,034
EBITDA (USD Mil)	1,000	1,064	1,005	1,024	1,113	1,134
Net Income (USD Mil)	1,059	1,114	1,070	1,089	1,170	1,189
Earnings Per Share (Diluted) (USD)	8.24	8.78	9.15	9.45	11.21	11.70
Adjusted Earnings Per Share (Diluted) (USD)	10.11	10.68	11.30	11.66	13.68	14.27
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 30 Jul 2026

Cost of Equity %	8.0
Pre-Tax Cost of Debt %	5.1
Weighted Average Cost of Capital %	7.5
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	5.0
Stage II Investment Rate %	30.0
Perpetuity Year	15

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 30 Jul 2026

	USD Mil
Present Value Stage I	6,108
Present Value Stage II	2,414
Present Value Stage III	9,215
Total Firm Value	17,737
Cash and Equivalents	3,015
Debt	1,972
Other Adjustments	0
Equity Value	18,780
Projected Diluted Shares	100
Fair Value per Share (USD)	201.00

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Income Statement (USD)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (Mil)	2,415	2,565	2,725	2,818	3,023	3,284	3,561	3,834
Cost of Goods Sold (Mil)	283	319	362	401	388	398	410	418
Gross Profit (Mil)	2,132	2,246	2,364	2,416	2,635	2,885	3,151	3,415
Selling, General, Administrative & Other Expenses (Mil)	117	112	129	133	143	155	168	181
Advertising & Marketing Expenses (Mil)	747	863	947	1,009	1,077	1,164	1,256	1,346
Research & Development (Mil)	369	395	457	372	508	552	598	644
Depreciation & Amortization (if reported separately) (Mil)	0	0	0	0	0	0	0	0
Adjusted Operating Income (Mil)	899	876	831	902	907	1,014	1,128	1,244
Financial Non-Cash (Gains)/Losses (Mil)	0	0	0	0	0	0	1	1
Irregular Cash (Gains)/Losses (Mil)	0	0	0	0	0	0	0	0
Operating Income (Mil)	899	876	830	902	906	1,014	1,128	1,244
Net Interest Expense (Mil)	-77	-96	-114	-114	-114	-114	-114	-114
Income Tax Expense (Mil)	135	126	-112	152	153	169	186	204
After-Tax Items (Mil)	0	0	0	0	0	0	0	0
(Minority Interest) (Mil)	0	0	0	0	0	0	0	0
Net Income (Mil)	840	846	1,056	863	867	958	1,055	1,154
Adjusted Net Income (Mil)	997	1,039	1,306	1,059	1,070	1,170	1,275	1,381
Weighted Average Diluted Shares Outstanding (Mil)	118	113	110	105	95	85	77	69
Diluted Earnings Per Share	7.10	7.46	9.61	8.24	9.15	11.21	13.72	16.73
Diluted Adjusted Earnings Per Share	8.43	9.16	11.89	10.11	11.30	13.68	16.58	20.02
Dividends Per Common Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EBITDA (Mil)	957	972	932	999	1,003	1,111	1,226	1,343
Adjusted EBITDA (Mil)	957	972	934	1,000	1,005	1,113	1,228	1,344

Accounts Payable = Accounts Payable + Deferred Income

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Key Cash Flow Items (USD)

Fiscal Year, ends 31 Dec	Actual			Forecast as of 30 Jul				
	2023	2024	2025	2026	2027	2028	2029	2030
Cash from Working Capital (K)	60,980	805	-90,850	338,819	-147,078	-24,548	24,543	-40,921
(Capital Expenditures) (K)	-18,600	-24,200	-26,600	-27,500	-29,509	-32,048	-34,758	-37,418
Depreciation (K)	23,100	24,000	24,800	25,639	27,513	29,880	32,406	34,886
Amortization (K)	34,700	71,800	77,200	71,831	69,371	67,806	66,185	64,126
Net New (Investment), Organic (K)	100,180	72,405	-15,450	408,790	-79,704	41,089	88,376	20,673
(Purchases)/Sales of Companies & Assets (K)	-458,800	-185,800	-273,100	-25,000	-25,000	-25,000	-25,000	-25,000
Net New (Investment), Total (K)	-358,620	-113,395	-288,550	383,790	-104,704	16,089	63,376	-4,327
Other Non-Cash Items, From Cash Flows (K)	919,120	955,795	1,188,250	0	0	0	0	0
Free Cash Flow to the Firm (K)	1,334,909	1,604,496	1,828,703	1,150,395	666,091	878,058	1,022,453	1,053,269

Balance Sheet (USD)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Assets								
Cash and Equivalents (Mil)	1,530	1,372	3,015	2,854	2,183	1,735	1,439	1,175
Inventory (Mil)	0	0	0	0	0	0	0	0
Accounts Receivable (Mil)	658	729	769	788	853	923	1,000	1,079
Net Property, Plant and Equipment (Mil)	80	81	83	69	56	44	32	21
Goodwill (Mil)	1,554	1,696	1,904	1,914	1,924	1,934	1,944	1,954
Other Intangibles (Mil)	194	201	214	173	133	95	58	22
Other Operating Assets (Mil)	1,679	1,676	1,821	1,821	1,821	1,821	1,821	1,821
Non-Operating Assets (Mil)	0	0	0	0	0	0	0	0
Total Assets (Mil)	5,696	5,755	7,806	7,619	6,970	6,553	6,295	6,073
Liabilities								
Accounts Payable (Mil)	1,462	1,526	1,558	1,907	1,791	1,815	1,902	1,926
Debt (Mil)	0	0	1,972	1,972	1,972	1,972	1,972	1,972
Other Operating Liabilities (Mil)	1,412	1,439	1,395	1,395	1,395	1,395	1,395	1,395
Non-Operating Liabilities (Mil)	0	0	0	0	0	0	0	0
Total Liabilities (Mil)	2,875	2,965	4,924	5,274	5,158	5,182	5,269	5,293
Equity								
Shareholders' Equity (Mil)	2,821	2,789	2,882	2,345	1,812	1,371	1,026	780
Minority Interest (Mil)	0	0	0	0	0	0	0	0
Total Equity (Mil)	2,821	2,789	2,882	2,345	1,812	1,371	1,026	780

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ESG Risk Rating Breakdown

Exposure

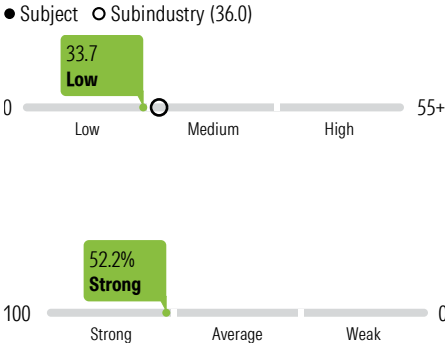
Company Exposure ¹	33.7
- Manageable Risk	31.3
Unmanageable Risk²	2.4

Management

Manageable Risk	31.3
- Managed Risk ³	16.4
Management Gap⁴	15.0

Overall Unmanaged Risk

17.3



- ▶ Exposure represents a company's vulnerability to ESG risks driven by their business model
- ▶ Exposure is assessed at the Subindustry level and then specified at the company level
- ▶ Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure
- ▶ Management measures a company's ability to manage ESG risks through its commitments and actions
- ▶ Management assesses a company's efficiency on ESG programs, practices, and policies
- ▶ Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Rating is of Jul 31, 2026. Highest Controversy Level is as of Jul 08, 2026. Sustainalytics Subindustry: Enterprise and Infrastructure Software. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 52.2% 4. Management Gap assesses risks that are not managed, but are considered manageable

Peer Analysis 31 Jul 2026

Company Name	Exposure	Management	ESG Risk Rating
Check Point Software Technologies Ltd	33.7 Low 0 —●— 55+	52.2 Strong 100 —●— 0	17.3 Low 0 —●— 40+
Cisco Systems Inc	36.4 Medium 0 —●— 55+	70.7 Strong 100 —●— 0	12.4 Low 0 —●— 40+
Allot Ltd	38.1 Medium 0 —●— 55+	25.6 Average 100 —●— 0	29.0 Medium 0 —●— 40+
Fortinet Inc	32.8 Low 0 —●— 55+	56.1 Strong 100 —●— 0	15.7 Low 0 —●— 40+
Palo Alto Networks Inc	28.7 Low 0 —●— 55+	63.7 Strong 100 —●— 0	11.5 Low 0 —●— 40+

Morningstar Sustainalytics ESG ratings are issued in accordance with the EU ESG Ratings Regulation (2024/3005). For regulatory disclosures related to Morningstar Sustainalytics ESG ratings, please see [Morningstar Sustainalytics Ratings Transparency](#).

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our single-point star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural

feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in working capital accounts, and capital spending. Based on

these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in as-

Morningstar Equity Research Star Rating Methodology



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signing a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

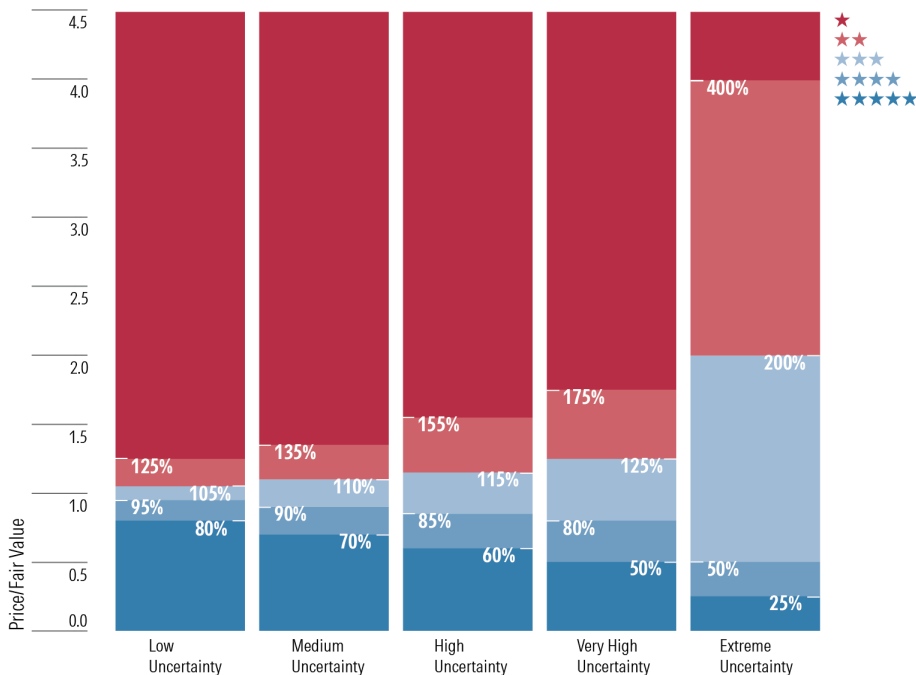
Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

Morningstar Equity Research Star Rating Methodology



For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other

factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

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Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score. Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

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